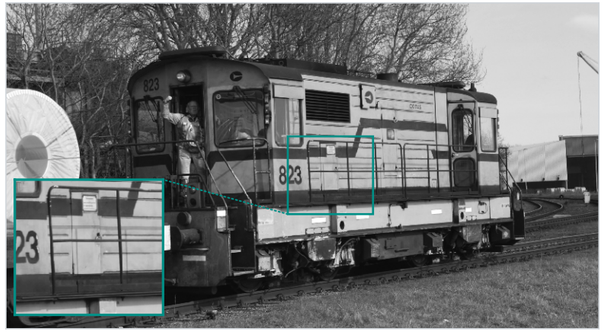
Single-plane depth transfer with a frozen HoloNet

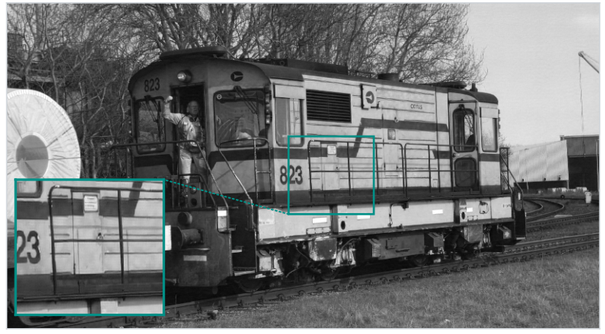
No additional training | Same target, pixel pitch and output coordinates

To 18 cm

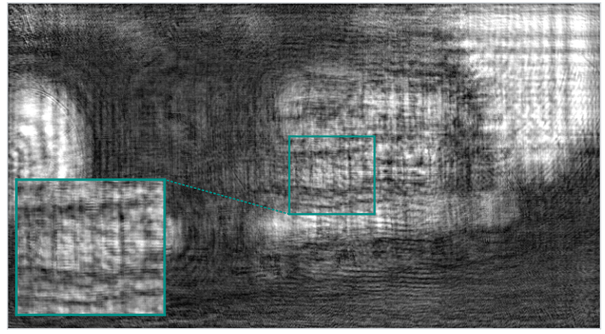
Target amplitude



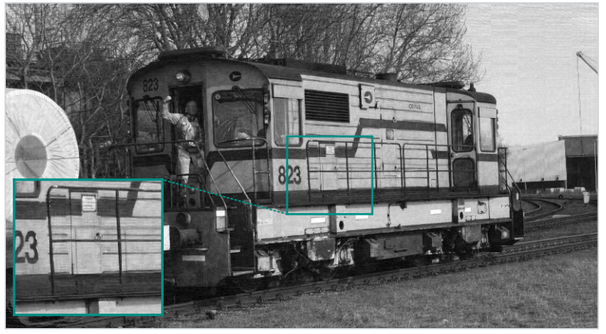
Original model at 20 cm
31.13 dB (reference)



Change distance only
13.84 dB

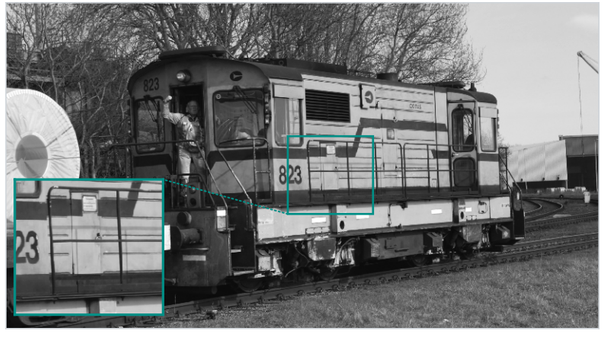


Carrier-aligned transfer
29.13 dB

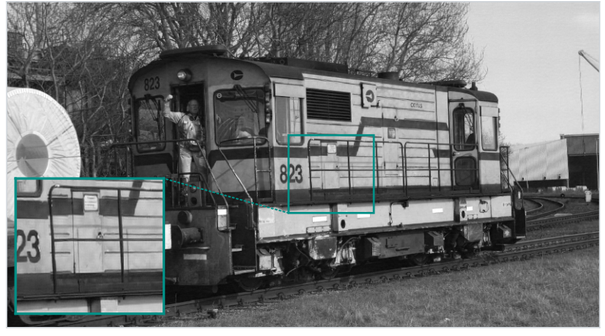


To 22 cm

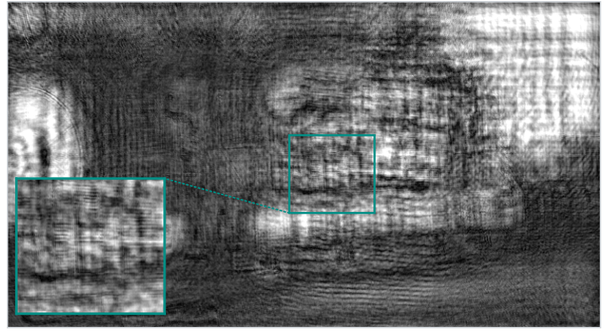
Target amplitude



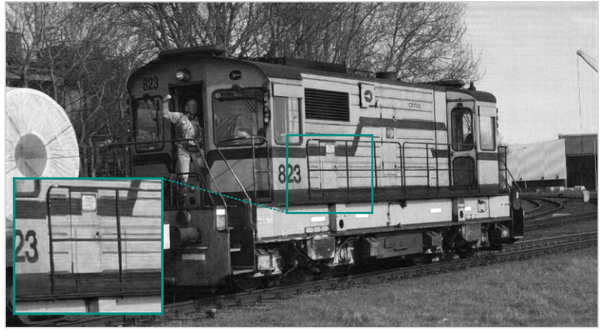
Original model at 20 cm
31.13 dB (reference)



Change distance only
13.78 dB



Carrier-aligned transfer
29.38 dB



Amplitude PSNR on the same 1600 x 880 ROI, with fitted brightness. The 20 cm column is a single-depth reference.
All panels use the saved amplitude scale; identical green boxes locate the enlarged details. One image; simulation only.

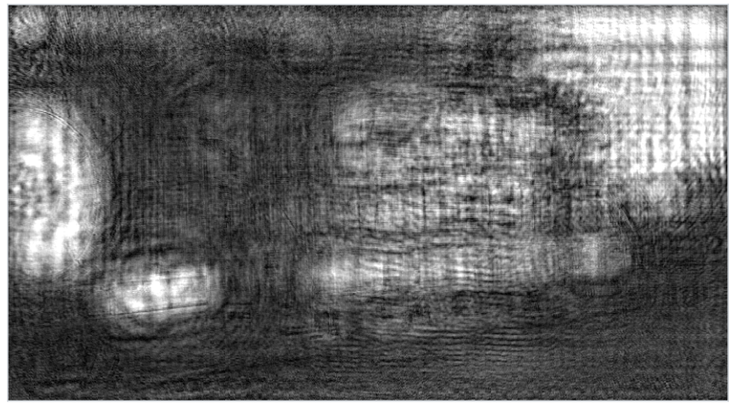
One phase-only hologram, two reconstruction depths

18 cm / 22 cm | Frozen initial-phase and final encoding networks | No additional training

Desired content / target
18 cm layer mask



Without carrier alignment
13.61 dB (assigned region)



With carrier alignment
25.92 dB (assigned region)

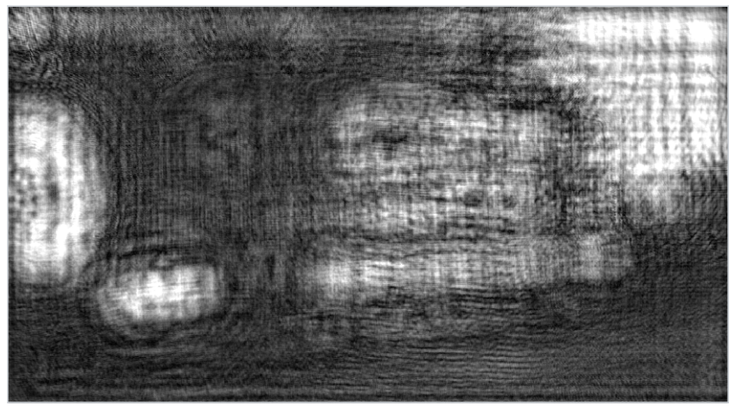


18 cm plane

22 cm layer mask



14.17 dB (assigned region)



24.58 dB (assigned region)

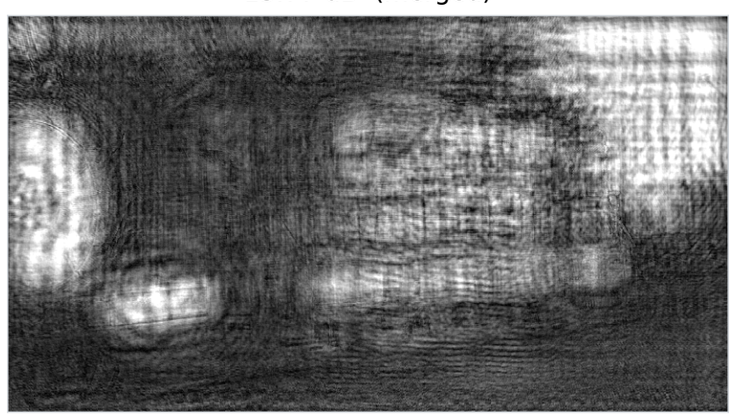


22 cm plane

Full target



13.77 dB (merged)



25.47 dB (merged)



Depth-masked merge

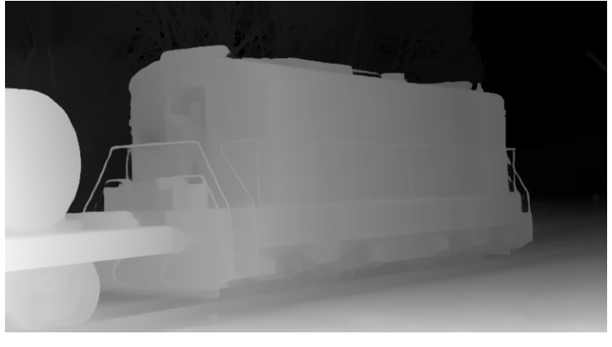
Each reconstruction column comes from ONE hologram; the last row selects pixels by the depth mask.
Black areas in the target layer mark unscored pixels, not required darkness. Defocused light remains visible in the complete planes.
One fitted brightness factor is shared by both planes and the merge within each method.

Input geometry and fixed two-depth assignment

Target amplitude



Depth guide (normalized)



18 cm: depth > 0.1
70.1% of ROI pixels



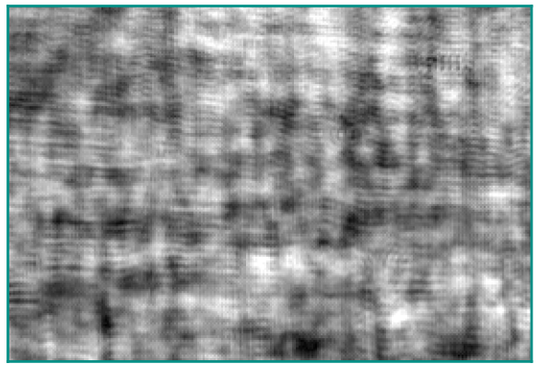
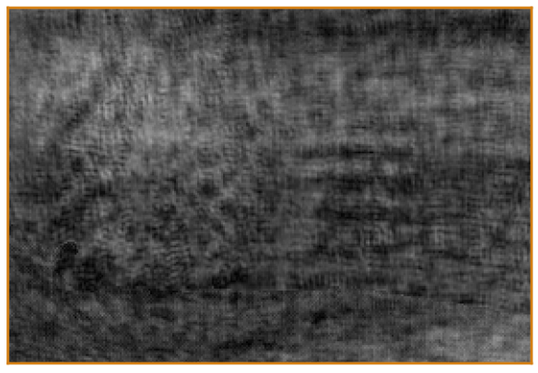
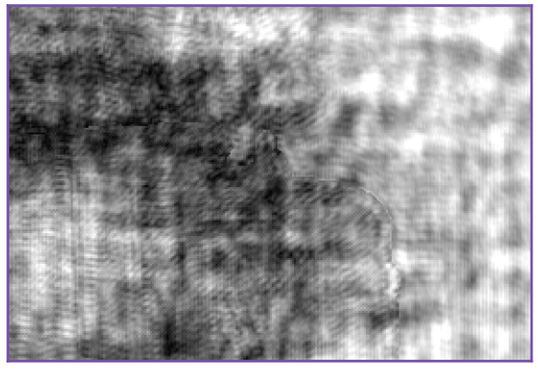
22 cm: depth <= 0.1
29.9% of ROI pixels



RGB and depth use the same aspect-preserving resize/crop and padding as the recorded experiments.
The guide supplies a binary assignment; its values are not interpreted as metric depth.

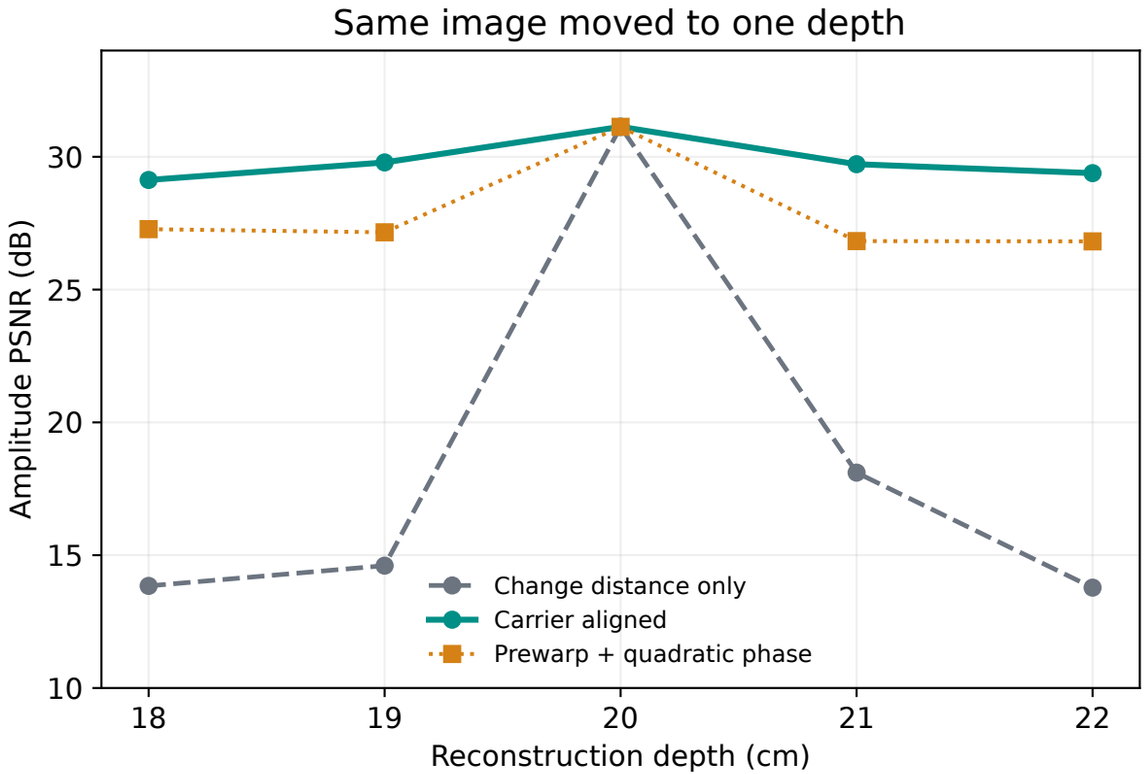
Local details at fixed pixel coordinates

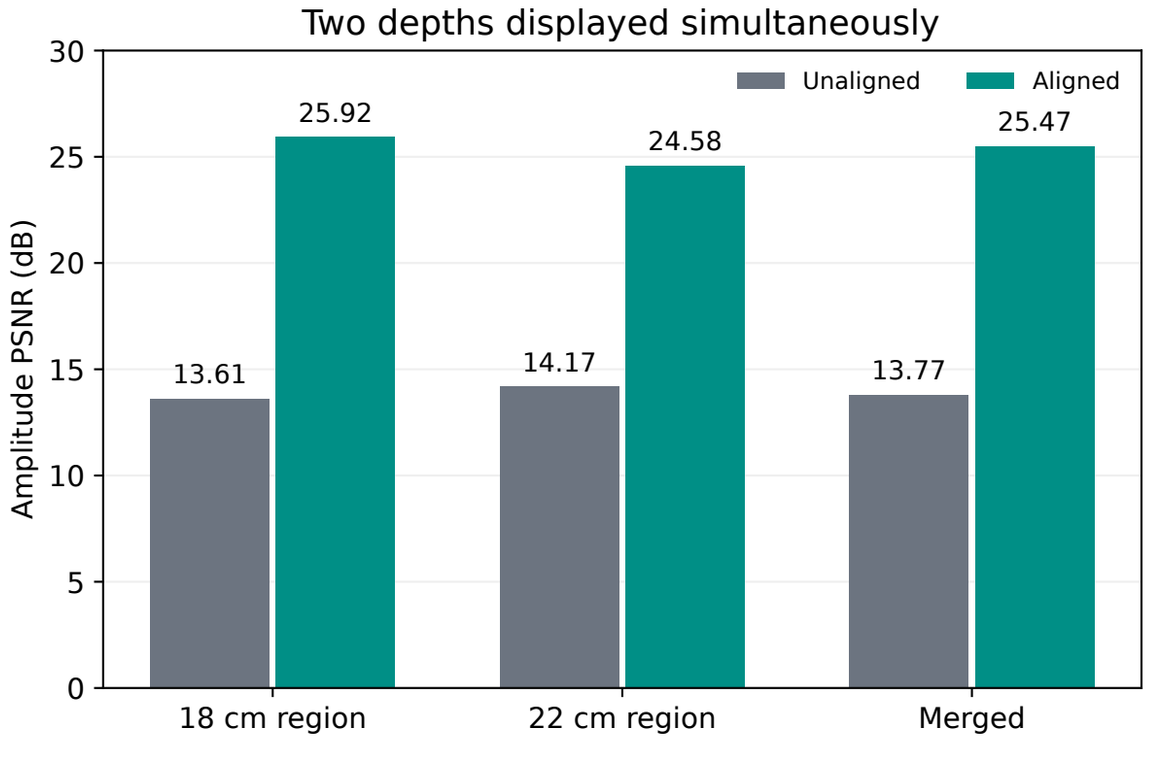
No local contrast normalization or sharpening; nearest-neighbor display

	Target	Two-depth: unaligned	Two-depth: aligned	Single-depth 20 cm (reference task)
Near: locomotive grille				
Far: fine branches				
Depth boundary: roof edge				

The aligned two-depth result still shows boundary ringing and residual texture. The 20 cm result is contextual, not a two-depth reconstruction. Exact crop coordinates and source hashes are recorded in manifest.json.

Quantitative validation without additional training





Metrics use original float records. At 20 cm the lens transform is identity; its recorded baseline is shown.
Single-image simulation only. The two panels are different tasks; neither is directly comparable with PHAROS dataset scores.